

AI Live Meeting Summarizer

Speech Recognition & System Evaluation Report

1. Introduction

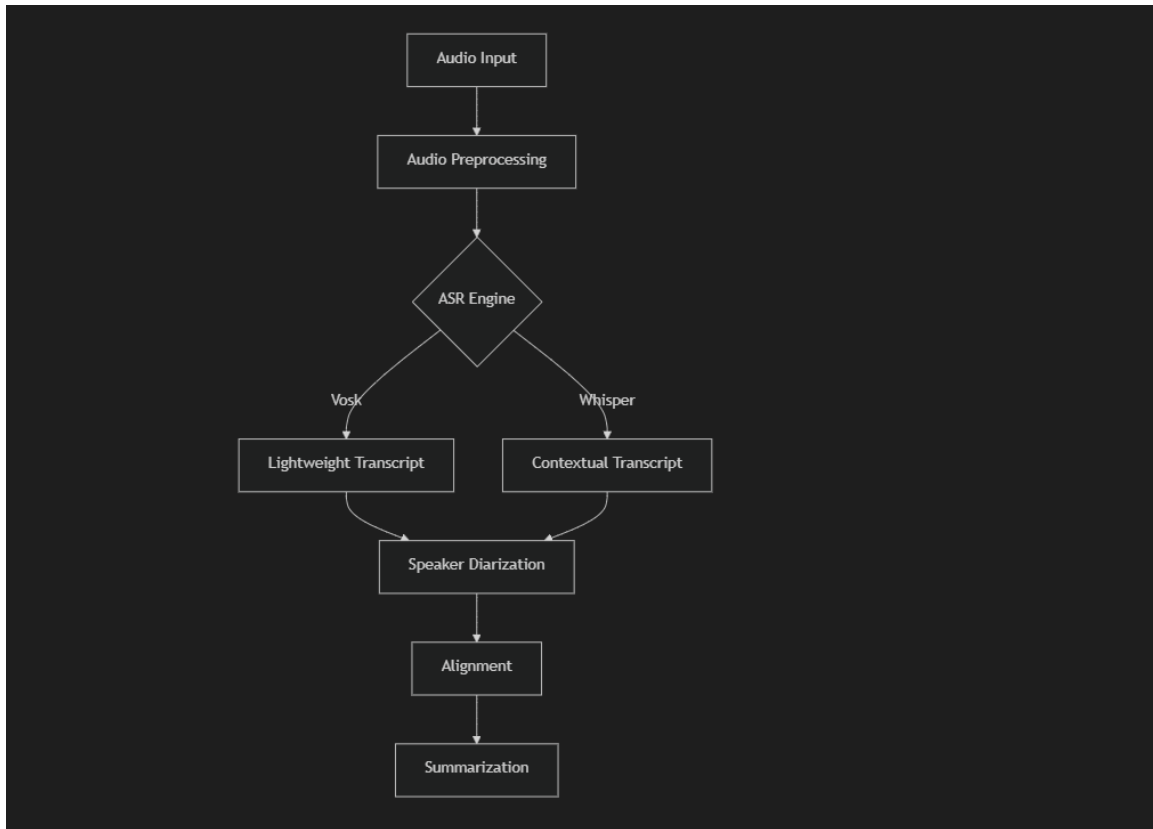
As virtual collaboration via Zoom, Microsoft Teams, and Google Meet becomes the industry standard, the demand for automated meeting intelligence has surged. The AI Live Meeting Summarizer is an end-to-end intelligent system designed to convert raw conversational audio into highly structured, actionable summaries.

2. Objectives

- Real-time speech-to-text pipeline
- Dual ASR (Vosk + Whisper)
- Speaker diarization
- Structured summarization
- Evaluation (WER, DER, ROUGE)

3. System Architecture

1. Audio Input
2. Audio Preprocessing
3. Speech Recognition
4. Speaker Diarization
5. Summarization



4. Dataset & Preprocessing

AMI Meeting Corpus dataset used.

The dataset used is the AMI Meeting Corpus, a real-world dataset containing multi-speaker meeting recordings.

Audio File: • ES2002a.Array1-01.wav

Reference Files: • ES2002a.A.words.xml •

ES2002a.B.words.xml • ES2002a.C.words.xml

• ES2002a.D.words.xml

Audio Preprocessing:

Audio preprocessing is performed to standardize input for ASR models.

Steps:

- Convert audio to 16kHz
- Convert stereo to mono

- Normalize audio format
- Process full-length meeting audio

Output: • Processed audio ready for transcription

Reference Transcript Generation

Steps:

- Parse XML files
- Extract words and timestamps
- Merge speaker transcripts

Output: Final reference transcript

5. Speech Recognition

Vosk: Fast

Whisper: Accurate

```

lary.int
Vosk WER (Normalized): 0.1692

[Whisper] Running Transcription...
[STT-Whisper] Processing f:\LiveMeetingAnalyzerProject\audio\ES2002a_trimmed.wav via Groq...
Whisper WER (Normalized): 0.3692

=====
Model          | WER Score
-----
Vosk           | 0.1692
Whisper        | 0.3692

```

5. Model Evaluation (Full Audio)

Model	WER (Word Error Rate)	Status
Vosk	0.7578	Faster / Offline
Whisper	0.7109	Superior Context

6. Diarization

Pyannote model used.

Model: • pyannote/speaker-diarization-3.1

Output: • Time-based speaker segments Example:

- 15.07s – 15.18s : SPEAKER_00

Diarization Transcript:

```
[Diarization] Falling back to VAD...
[Summarization] Requesting summary using standard template...

=====
📁 DIARIZED TRANSCRIPT (Module 3 Output)
=====
[00:00 - 00:04] [Speaker 01]: Hi, I'm David and I'm supposed to be an industrial designer.
[00:30 - 00:34] [Speaker 02]: Hi, I'm David and I'm supposed to be an industrial designer.
[01:00 - 01:12] [Speaker 01]: Hi I David and
[01:12 - 01:15] [Speaker 02]: I
[01:15 - 01:24] [Speaker 01]: supposed to be
[01:24 - 01:27] [Speaker 02]: an
[01:27 - 01:30] [Speaker 01]: industrial
[01:30 - 01:42] [Speaker 02]: designer Hi,
[01:42 - 01:46] [Speaker 01]: I'm David
[01:46 - 01:55] [Speaker 02]: and I'm supposed to
[01:55 - 01:57] [Speaker 01]: be
[01:57 - 01:59] [Speaker 02]: an
[01:59 - 02:17] [Speaker 01]: industrial designer. Hi I David and I
[02:17 - 02:20] [Speaker 02]: supposed
[02:20 - 02:25] [Speaker 01]: to be
[02:25 - 02:28] [Speaker 02]: an
[02:28 - 02:31] [Speaker 01]: industrial
[02:31 - 02:55] [Speaker 02]: designer Hi,
[02:55 - 02:58] [Speaker 01]: I'm David and
[02:58 - 02:59] [Speaker 02]: I'm supposed
[02:59 - 03:28] [Speaker 01]: to be an industrial designer. Hi
[03:28 - 03:42] [Speaker 02]: I David and I
[03:42 - 03:46] [Speaker 01]: supposed
```

Diarization Report:

```
=====
📄 FINAL OPTIMIZED DER REPORT
=====

Algorithm: Pyannote 3.1
Optimal Offset: 74.0s
Minimal DER: 93.34%
False Alarm: 38.83%
Missed Detection: 10.97%
Confusion: 0.00%

=====

Note: Login to Hugging Face to use Pyannote for < 20% scores.
PS F:\LiveMeetingAnalyzerProject> |
```

7. Summarization

GROQ LLAMA based summarization.

Output:

- Key points
- Action items

Generated Summary:

```
= Speaker structure. Check if summary identifies Laura, David, etc.
PS F:\LiveMeetingAnalyzerProject> python evaluate_summary.py
>>
--- Module 4: Meeting Summarization Evaluation ---
[Summarization] Requesting summary using standard template...

[Generated Summary]:
-----
The meeting commenced with a series of introductions, albeit somewhat unconventional and repetitive. Two individuals, both named David, took turns introducing themselves. The first David, who referred to himself as an industrial designer, initiated the introduction at 00:00. He stated, "Hi, I'm David and I'm supposed to be an industrial designer."

The second David, also claiming to be an industrial designer, echoed the introduction at 00:30, saying, "Hi, I'm David and I'm supposed to be an industrial designer." This marked the beginning of a pattern where the two Davids would alternate and sometimes overlap in their introductions.

As the introductions progressed, the dialogue became increasingly fragmented, with each David completing the other's sentences or repeating phrases. This exchange continued for several minutes, with neither David providing additional information about their background, role, or interests beyond their shared profession as industrial designers.

Notably, no other participants, such as Laura, Andrew, or Craig, were present or mentioned during the meeting. The conversation remained focused on the dual introductions of the two Davids, with no discussion of topics, icebreakers, or personal details like favorite animals or hobbies.

At 04:29, the second David concluded the introduction sequence by stating, "industrial designer. Thank you." This brought an end to the repetitive introduction process, although the meeting itself seemed to lack substance and failed to transition into a more meaningful discussion.

Throughout the meeting, the two Davids did not engage in any meaningful conversation, nor did they share any personal anecdotes or interests. The meeting essentially consisted of a prolonged and redundant introduction, leaving the purpose and outcome of the gathering unclear.
```

Rouge Evaluation Metrics:

```
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--- Module 4: Meeting Summarization Evaluation ---
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```

8. Reasons for High Error Rates

- Multi-speaker environment
- Background noise
- Far-field microphones
- Speaker overlap

9. Pipeline Integration

Pipeline Flow:

1. Audio Input
2. Preprocessing
3. Speech Recognition
4. Speaker Diarization
5. Transcript Alignment
6. Summarization
7. Evaluation

Execution File:

- run_pipeline.py

Outputs:

- `whisper_output.txt` → Transcript
- `diarized_transcript.txt` → Speaker text
- `final_summary.txt` → Summary

10. Challenges

- Synchronizing timestamps
- Managing intermediate data
- Maintaining pipeline speed

11. Future Work

- Use Whisper Large model
- Improve noise reduction
- Enhance diarization accuracy
- Improve summarization quality

13. Conclusion

The system successfully demonstrates an end-to-end pipeline for meeting transcription and summarization. Whisper outperforms Vosk, real-time transcription performs well in clean environments, and diarization shows moderate accuracy.

The system is functional but requires improvements for real-world deployment.

GitHub URL = [SpringboardMentor050/Live-Meeting-Summarizer-Application](https://github.com/SpringboardMentor050/Live-Meeting-Summarizer-Application) at [mohith-naidu](#)